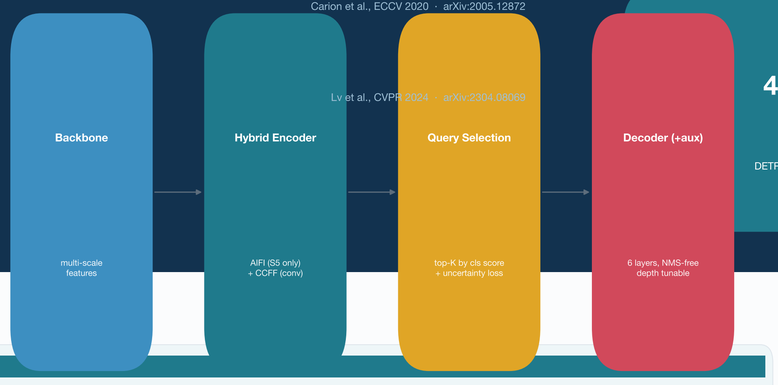
# NMS-Free Real-Time Detection:  DETR -> RT-DETR

Why moving duplicate suppression from post-processing into the loss buys accuracy and speed at once

Carion et al., ECCV 2020 · arXiv:2005.12872

Lv et al., CVPR 2024 · arXiv:2304.08069




**42.0**
DETR AP (41M)

**53.1**
RT-DETR AP

**108**
FPS on T4 (FP16)

## 1 The NMS dilemma

Two thresholds move accuracy and time in opposite directions

conf 0.001 -> 0.05:  AP 52.9 -> 51.2,  NMS 2.36 -> 1.06 ms

IoU 0.5 -> 0.8:  AP 52.1 -> 52.8,  NMS 2.24 -> 2.46 ms

## 2 Three moves

One-to-one matching kills duplicates at training time -> no NMS at inference

8400 vs 400 tokens = 441x attention gap -> keep attention only on S5

1) Duplicates -> loss: optimal bipartite matching (one slot per GT)

matching cost = probability, loss = log-likelihood

2) Compute -> design: encoder was 49% of GFLOPs, 11% of AP

AIFI (S5 only, 400 tokens) + CCFF (conv fusion)

138% / 120% are relative counts, not percentage-point gains

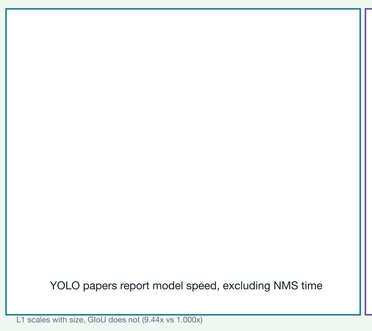
| Model | AP | AP_S | AP_L | Params | FPS |
|---|---|---|---|---|---|
| DETR (R50) | 42.0 | 20.5 | 61.1 | 41M | 28 |
| **Accuracy and speed together** | | | | | |
| Faster R-CNN-FPN+ | 42.0 | 26.6 | 53.4 | 42M | 26 |
| **RT-DETR (R50)** | 53.1 | 34.8 | 70.0 | **42M** | 108 |

DETR: same params as Faster R-CNN-FPN+, but AP_S -5.5 / AP_L +7.8 | RT-DETR: +2.2 AP and 21x faster than DINO-Deformable (5 -> 108 FPS)

Conditional comparison: 640x640 / 72 ep (RT-DETR) vs 800x1333 / 500 ep (DETR)

Fig. 2  Accuracy vs speed (paper Table 2): RT-DETR is top-right

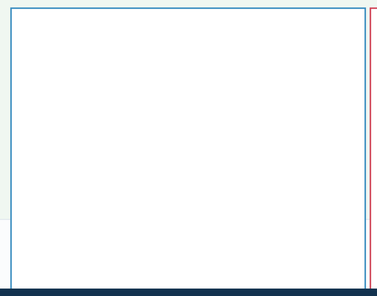
## 4 Our own experiments



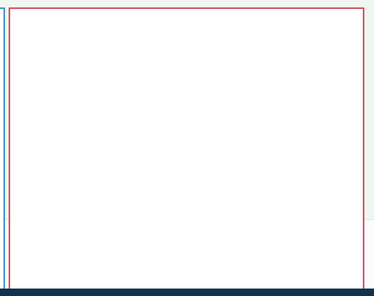

L1 scales with size, GIoU does not (9.44x vs 1.000x)



Attention cost ~ M^2 (42.3x for 8x tokens)



Encoder variant latency order matches the paper



Uncertainty alone has a degenerate optimum

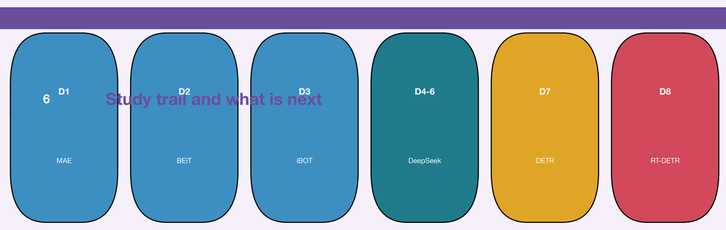
## 5 Limitations we kept honest

Small objects still weak: AP_S 34.8 (-0.5 vs YOLOv8-L; R101 -0.9)

box loss = scale-invariant GIoU + L1 (weights 2 / 5)

49% GFLOPs / 11% AP is a second-hand statistic, not our measurement

## 6 Study trail and what is next




16 deep-dive docs, 4 runnable demos, 22 charts, 0 citation errors

**Move duplicate removal into the loss, move compute into the design, move initialization into quality.**

arXiv:2005.12872 · arXiv:2304.08069 · code: github.com/facebookresearch/detr, github.com/lyuwenyu/rt-detr · figures: our Day 07-08 measurements (synthetic, CPU)

**set prediction**
Next: GLIP + SAM (open-vocab detection -> prompt segmentation)

**hybrid encoder**

**uncertainty-minimal**

Then: CLIP -> LLaVA (2 days) -> VL-JEPA | parallel 3D track